Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The complete solution set of the inequality $\cos^{-1}(\cos 4) > 3x^2 - 4x$ is

(A) $\left(0, \frac{2+\sqrt{6\pi-8}}{3}\right)$

(B) $\left(\frac{2-\sqrt{6\pi-8}}{3}, 0\right)$

(C) $(-2, 2)$

(D) $\left(\frac{2-\sqrt{6\pi-8}}{3}, \frac{2+\sqrt{6\pi-8}}{3}\right)$

Q.2

The value of
$\tan^{-1}\left(\frac{1}{2}(\tan 2\ \mathrm{A}) + \tan^{-1}(\cot \mathrm{A}) + \tan^{-1}\left(\cot^3\ \mathrm{A}\right)\right)$ is

(A) 0 if $\frac{\pi}{4} < A < \frac{\pi}{2}$

(B) π, if $0 < \mathrm{A} < \frac{\pi}{4}$

(C) both (a) and (b)

(D) None of these

Q.3

If $\left|\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)\right| < \frac{\pi}{3}$, then

(A) $x \in \left[-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right]$

(B) $x \in \left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$

(C) $x \in \left(0, \frac{1}{\sqrt{3}}\right)$

(D) None of these

Q.4

The range of the function $f(x) = \sin^{-1}(\log[x]) + \log\left(\sin^{-1}[x]\right)$; (where [.] denotes the greatest integer function) is

(A) R

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(B) $[1, 2)$

(C) $\left\{\log \frac{\pi}{2}\right\}$

(D) $\{-\sin 1\}$

Q.5

If θ and ϕ are the roots of the equation $8x^2 + 22x + 5 = 0$, then

(A) both $\sin^{-1}\theta$ and $\sin^{-1}\phi$ are real

(B) both $\sec^{-1}\theta$ and $\sec^{-1}\phi$ are real

(C) both $\tan^{-1}\theta$ and $\tan^{-1}\phi$ are real

(D) None of these

Q.6

If $\cot^{-1}\frac{n}{\pi} > \frac{\pi}{6}, n \in \mathbf{N}$, then the maximum value of n is

(A) 4

(B) 5

(C) 6

(D) 8

Q.7

Which of the following is the principal value branch of $\operatorname{cosec}^{-1} x$?

(A) $\left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$

(B) $(0, \pi) - \left[\frac{\pi}{2}\right]$

(C) $\left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$

(D) $\left[\frac{-\pi}{2}, \frac{\pi}{2}\right] - \{0\}$

Q.8

The solution set of the equation
$\tan^{-1} x - \cot^{-1} x = \cos^{-1}(2 - x)$ will lie in the
interval

(A) $[0, 1]$

(B) $[-1, 1]$

(C) $[1, 3]$

(D) None of these

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Q.9

Complete solution set of $\tan^2\left(\sin^{-1}x\right) > 1$ is

(A) $\left(-1, -\frac{1}{\sqrt{2}}\right) \cup \left(\frac{1}{\sqrt{2}}, 1\right)$

(B) $\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) \sim \{0\}$

(C) $(-1, 1) \sim \{0\}$

(D) None of these

Q.10

Let $x \in (0, 1)$. The set of all x such that $\sin^{-1}x > \cos^{-1}x$, is the interval:

(A) $\left(\frac{1}{2}, \frac{1}{\sqrt{2}}\right)$

(B) $\left(\frac{1}{\sqrt{2}}, 1\right)$

(C) $(0, 1)$

(D) $\left(0, \frac{\sqrt{3}}{2}\right)$

Q.11

The range of $f(x) = \sin^{-1}x + \tan^{-1}x + \sec^{-1}x$
is

(A) $\left(\frac{\pi}{4}, \frac{3\pi}{4}\right)$

(B) $\left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$

(C) $\left\{\frac{\pi}{4}, \frac{3\pi}{4}\right\}$

(D) None of these

Q.12

The equation $\sin^{-1}\left(3x - 4x^3\right) = 3\sin^{-1}(x)$ is true
for all values of $\mathbf{x}$ lying in which one of the following intervals?

(A) $\left[-\frac{1}{2}, \frac{1}{2}\right]$

(B) $\left[\frac{1}{2}, 1\right]$

(C) $\left[-1, -\frac{1}{2}\right]$

(D) $[-1, 1]$

Q.13

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What is the value of $\sec^2 \tan^{-1}\left(\frac{5}{11}\right)$?

(A) $121/96$

(B) $211/921$

(C) $146/121$

(D) $267/121$

Q.14

$\tan^{-1}\left[\frac{\cos x}{1+\sin x}\right] =$

(A) $\frac{\pi}{4} - \frac{x}{2}$

(B) $\frac{\pi}{4} + \frac{x}{2}$

(C) $\frac{x}{2}$

(D) $\frac{\pi}{4} - x$

Q.15

$\tan^{-1}\left(\frac{1-x}{1+x}\right), 0 \leq x \leq 1$ The smallest and the largest values of $\tan^{-1}\left(\frac{1-x}{1+x}\right) =$

(A) $0, \pi$

(B) $0, \frac{\pi}{4}$

(C) $-\frac{\pi}{4}, \frac{\pi}{4}$

(D) $\frac{\pi}{4}, \frac{\pi}{2}$

Answers & Solutions

Q.1 Answer:

$\frac{2 - \sqrt{6\pi - 8}}{3}, \frac{2 + \sqrt{6\pi - 8}}{3}$

Solution:

As, $\cos^{-1}(\cos 4) = \cos^{-1}\cos(2\pi - 4) = 2\pi - 4$

$\Rightarrow \quad 2\pi - 4 > 3x^2 - 4x$

$\Rightarrow \quad 3x^2 - 4x - 2\pi - 4 < 0$

$\Rightarrow \quad \frac{2 - \sqrt{6\pi - 8}}{3} < x < \frac{2 + \sqrt{6\pi - 8}}{3}$

Q.2 Answer:

both (a) and (b)

Solution:

We know that $\cot A > 1$ **if** $0 < A < \frac{\pi}{4}$

and $\cot A < 1$ **if** $\frac{\pi}{4} < A < \frac{\pi}{2}$

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$\tan^{-1}(\cot A)+\tan^{-1}\left(\cot^3 A\right)=\pi+\tan^{-1}\frac{\cot A+\cot^3 A}{1-\cot^4 A}$

If $0<A<\frac{\pi}{4}$

and $\tan^{-1}(\cot A)+\tan^{-1}\left(\cot^3 A\right)$

$=\tan^{-1}\frac{\cot A+\cot^3 A}{1-\cot^4 A}$ **if** $\frac{\pi}{4}<A<\frac{\pi}{2}$

Also, $\frac{\cot A+\cot^3 A}{1-\cot^4 A}=\frac{\cot A\cos ec^2 A\cdot\sin^4 A}{\sin^4 A-\cos^4 A}$

$=\frac{\sin A\cos A}{(\sin^2 A+\cos^2 A)(\sin^2 A-\cos^2 A)}$

$=-\frac{\sin 2A}{2\cos 2A}=-\frac{1}{2}\tan 2A$

Hence, $\tan^{-1}\left(\frac{1}{2}\tan 2A\right)+\tan^{-1}(\cot A)+\tan^{-1}\left(\cot^3 A\right)$

$=\begin{cases}\pi & \text{if } 0<A<\frac{\pi}{4}\\ 0 & \text{if } \frac{\pi}{4}<A<\frac{\pi}{2}\end{cases}$

$[$ **Since,** $\tan^{-1}(-x)=-\tan^{-1}x]$

Q.3 Answer:

$x\in\left(-\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}\right)$

Solution:

We have, $\left|\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)\right|<\frac{\pi}{3}$

$\Rightarrow-\frac{\pi}{3}<\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)<\frac{\pi}{3}$

$\Rightarrow 0\le\cos^{-1}\frac{1-x^2}{1+x^2}<\frac{\pi}{3}\Rightarrow\frac{1}{2}<\frac{1-x^2}{1+x^2}\le 1$

$\Rightarrow 1+x^2<2\left(1-x^2\right)\le 2\left(1+x^2\right)$

$\Rightarrow 0\le x^2<\frac{1}{3}\Rightarrow-\frac{1}{\sqrt{3}}<x<\frac{1}{\sqrt{3}}$

Q.4 Answer:

$\left\{\log\frac{\pi}{2}\right\}$

Solution:

$\sin^{-1}(\log[x])$ **is defined if** $-1\le\log[x]\le 1$ **and** $[x]>0$

$\Rightarrow\frac{1}{e}\le[x]\le e\Rightarrow[x]=1,2\Rightarrow x\in[1,3)$

Again, $\log\left(\sin^{-1}[x]\right)$ **is defined if**

$\sin^{-1}[x]>0$ **and** $-1\le[x]\le 1$

$\Rightarrow[x]>0$ **and** $-1\le[x]\le 1\Rightarrow 0<[x]\le 1$

$\Rightarrow x\in[1,2)$

$\therefore$ **Domain of** $f(x)=[1,2)$ **For** $1\le x<2,[x]=1$

$\therefore\quad f(x)=\sin^{-1}0+\log\frac{\pi}{2}=\log\frac{\pi}{2},\forall x\in[1,2)$

$\therefore$ **Range of** $f(x)=\left\{\log\frac{\pi}{2}\right\}$

Q.5 Answer:

both $\tan^{-1}\theta$ and $\tan^{-1}\phi$ are real

Solution:

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$8x^2 + 22x + 5 = 0 \Rightarrow x = -\frac{1}{4}, -\frac{5}{2}$
$\because -1 < -\frac{1}{4} < 1$ **and** $-\frac{5}{2} < -1$
$\therefore \quad \sin^{-1}\left(-\frac{1}{4}\right)$ **exists but** $\sin^{-1}\left(-\frac{5}{2}\right)$ **does not exist**
$\sec^{-1}\left(-\frac{5}{2}\right)$ **exists but** $\sec^{-1}\left(-\frac{1}{4}\right)$ **does not exist,**
$\tan^{-1}\left(-\frac{1}{4}\right)$ **and** $\tan^{-1}\left(-\frac{5}{2}\right)$ **both exist**

Q.6 Answer:

5

Solution:

$\cot^{-1}\frac{n}{\pi} > \frac{\pi}{6} \Rightarrow \frac{\pi}{6} < \cot^{-1}\frac{n}{\pi} < \pi \Rightarrow -\infty < \frac{n}{\pi} < \sqrt{3}$
$\Rightarrow n < \pi\sqrt{3} (\because n > 0) \Rightarrow n \leq 5 (\because 5 < \pi\sqrt{3} < 6)$

Q.7 Answer:

$\left[\frac{-\pi}{2}, \frac{\pi}{2}\right] - \{0\}$

Solution:

Principal value branch of $\operatorname{cosec}^{-1} x = \left[\frac{-\pi}{2}, \frac{\pi}{2}\right] - \{0\}$

Q.8 Answer:

$[1, 3]$

Solution:

$\tan^{-1} x$ and $\cot^{-1} x$ exist for all $x \in R$
$\cos^{-1}(2 - x)$ exists if $-1 \leq 2 - x \leq 1$ i.e $1 \leq x \leq 3$
So, the given equation holds for $1 \leq x \leq 3$

Q.9 Answer:

$\left(-1, -\frac{1}{\sqrt{2}}\right) \cup \left(\frac{1}{\sqrt{2}}, 1\right)$

Solution:

$\tan^2\left(\sin^{-1} x\right) > 1$
$\Rightarrow \frac{\pi}{4} < \sin^{-1} x < \frac{\pi}{2}$ **or** $-\frac{\pi}{2} < \sin^{-1} x < -\frac{\pi}{4}$
$\Rightarrow x \in \left(\frac{1}{\sqrt{2}}, 1\right)$ **or** $x \in \left(-1, -\frac{1}{\sqrt{2}}\right)$
$\Rightarrow x \in \left(-1, -\frac{1}{\sqrt{2}}\right) \cup \left(\frac{1}{\sqrt{2}}, 1\right)$

Q.10 Answer:

$\left(\frac{1}{\sqrt{2}}, 1\right)$

Solution:

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Given $\sin^{-1} x > \cos^{-1} x$ where $x \in (0, 1)$
$\Rightarrow \sin^{-1} x > \frac{\pi}{2} - \sin^{-1} x \Rightarrow 2\sin^{-1} x > \frac{\pi}{2}$
$\Rightarrow \sin^{-1} x > \frac{\pi}{4} \Rightarrow x > \sin\frac{\pi}{4} \Rightarrow x > \frac{1}{\sqrt{2}}$
Maximum value of $\sin^{-1} x$ is $\frac{\pi}{2}$ So, maximum value of x is 1. So, $x \in \left(\frac{1}{\sqrt{2}}, 1\right)$.

Q.11 Answer:

$\left\{\frac{\pi}{4}, \frac{3\pi}{4}\right\}$

Solution:

$f(x) = \sin^{-1} x + \tan^{-1} x + \sec^{-1} x$; clearly,
the domain of $f(x)$ is $x = \pm 1$. Thus the range is $\{f(1), f(-1)\}$, i.e., $\left\{\frac{\pi}{4}, \frac{3\pi}{4}\right\}$.

Q.12 Answer:

$[-1, 1]$

Solution:

Let $\sin^{-1} \mathrm{x} = \theta \Rightarrow \mathrm{x} = \sin\theta$
$\sin^{-1}\left(3\sin\theta - 4\sin^3\theta\right) = \sin^{-1}\sin 3\theta$
$= 3\theta = 3\sin^{-1} x$
Equaion $\sin^{-1}\left(3x - 4x^3\right) = 3\sin^{-1} x$ is true for all
values of x lying in the interval $[-1, 1]$

Q.13 Answer:

$146/121$

Solution:

Let $\sec^2\left(\tan^{-1}\left(\frac{5}{11}\right)\right)$
$= 1 + \tan^2\left(\tan^{-1}\left(\frac{5}{11}\right)\right) \left(\because \sec^2\theta - \tan^2\theta = 1\right)$
$= 1 + \left[\tan\left(\tan^{-1}\left(\frac{5}{11}\right)\right)\right]^2 = 1 + \left(\frac{5}{11}\right)^2$
$= 1 + \frac{25}{121} = \frac{146}{121}$

Q.14 Answer:

$\frac{\pi}{4} - \frac{x}{2}$

Solution:

$$\tan^{-1}\left[\frac{\cos x}{1+\sin x}\right] = \tan^{-1}\left[\frac{\sin(\pi/2 - x)}{1+\cos(\pi/2 - x)}\right]$$
$$= \tan^{-1}\left[\frac{2\sin(\pi/4 - x/2)\cos(\pi/4 - x/2)}{2\cos^2(\pi/4 - x/2)}\right]$$
$$= \tan^{-1}\tan\left(\frac{\pi}{4} - \frac{x}{2}\right) = \frac{\pi}{4} - \frac{x}{2}$$

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Q.15 Answer:

$0, \frac{\pi}{4}$

Solution:

We have, $\tan^{-1}\left(\frac{1-x}{1+x}\right) = \tan^{-1} 1 - \tan^{-1} x = \frac{\pi}{4} - \tan^{-1} x$

Since $0 \leq x \leq 1 \Rightarrow 0 \leq \tan^{-1} x \leq \frac{\pi}{4}$

$\Rightarrow 0 \geq -\tan^{-1} x \geq \frac{-\pi}{4} \Rightarrow \frac{\pi}{4} \geq \frac{\pi}{4} - \tan^{-1} x \geq 0$

$\Rightarrow \frac{\pi}{4} \geq \tan^{-1}\left(\frac{1-x}{1+x}\right) \geq 0$

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